General Structure

Garcia-Suarez (2022) JMPS

$$\frac{d}{dx}\begin{bmatrix} f \\ f' \end{bmatrix} = \overbrace{\begin{bmatrix} 0 & A_{12} \\ A_{21} & 0 \end{bmatrix}}^{\boldsymbol{A} \in \mathfrak{sl}_2(\mathbb{R})} \begin{bmatrix} f \\ f' \end{bmatrix}$$

$$\rightarrow \boldsymbol{f}(x_{k+1}) = \overbrace{\exp(\boldsymbol{A}_k l)}^{\boldsymbol{T}_k(x_{k+1}, x_k) \in \boldsymbol{SL}_2(\mathbb{R})} \boldsymbol{f}(x_k)$$